Cysts

Dr.Manila

True Cysts – Definition & Origin

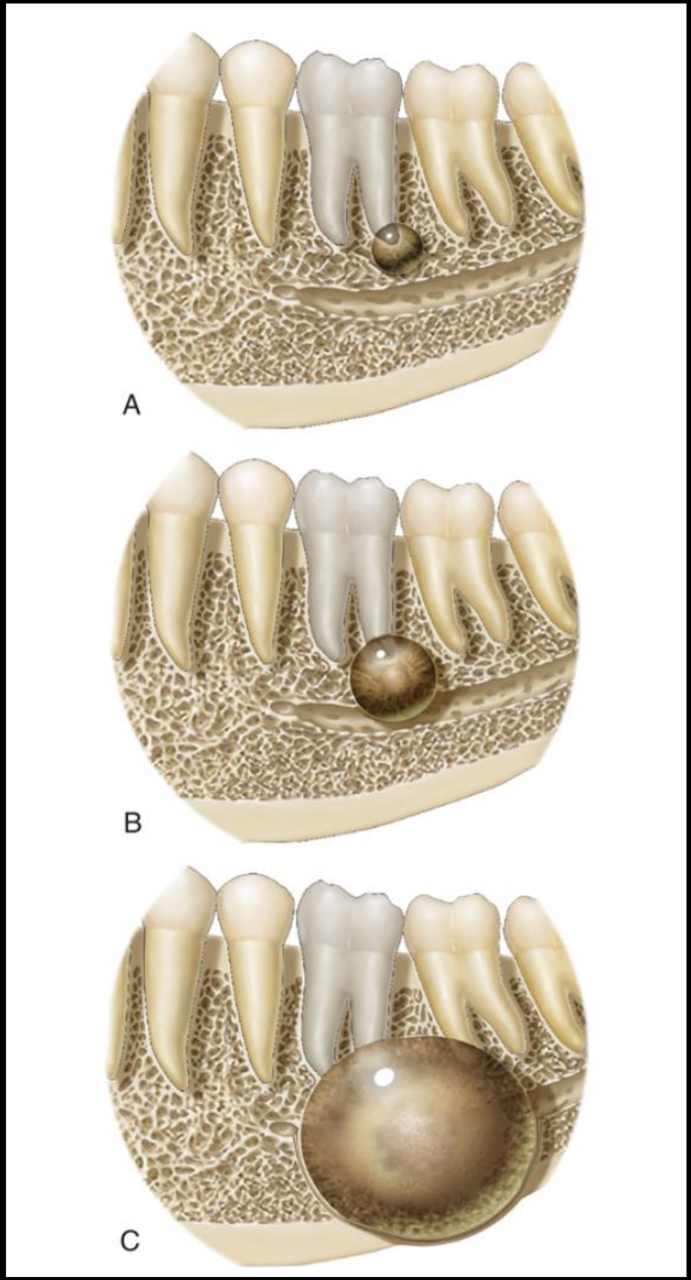
- True cyst: Epithelial lined pathologic cavity within bone
- Contains fluid from adjacent cells/tissues
- Epithelial origins:
    - **Odontogenic**: dental lamina, reduced enamel epithelium, rests of Malassez
    - **Nonodontogenic**: respiratory epithelium, fusion remnants
- Surrounded by thin connective tissue layer

Pseudocysts – Characteristics

- Lack epithelial lining or bone cavity
- Mimic radiologic features of true cysts
- Not true pathologic cavities

Disease Mechanism of True Cysts

- Initiated by epithelial cell proliferation
- Causes: genetic mutation or signaling molecules
- Osteoclast recruitment → bone resorption
- Debris accumulation → osmotic gradient
- Fluid influx → cavity expansion
- Connective tissue layer forms beneath epithelium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Clinical Features of Cysts

- Most common: firm to bony-hard swelling
- May be painful if:
  - Associated with nonvital tooth pulp
  - Secondarily infected
- **Role of Imaging in Diagnosis**
  - Reveals characteristic features for diagnosis
  - Assesses intra- and extraosseous involvement
  - Guides biopsy site selection

Imaging Features – Location & Periphery

- Common in maxillae/mandible; rare in condylar/coronoid processes
- Odontogenic cysts: tooth-bearing areas
- Nonodontogenic: antrum or soft tissues
- Well-defined periphery with thin radiopaque cortex
- Infected cysts: thicker, sclerotic cortex
- Scalloping around roots or cortex

Imaging Features – Shape & Internal Structure

- Smooth, curved "hydraulic" borders
- Uniformly radiolucent
- Long-standing cysts: cholesterol granules, calcifications
- Multilocular appearance due to differential epithelial proliferation
- False septation from scalloping

Effects on Surrounding Structures & Teeth

- Bone thins and expands smoothly
- Displacement of:
    - Inferior alveolar canal (inferior/lingual)
    - Maxillary sinus floor (superior)
- Tooth displacement and directional external resorption
- Loss of lamina dura and periodontal ligament space

Classification

- Odontogenic Cysts
  - Radicular cyst
  - Residual cyst
  - Dentigerous cyst
  - Odontogenic Keratocyst
  - Basal cell Nevus syndrome
  - Buccal Bifurcation cyst
  - Lateral periodontal cyst
  - Glandular odontogenic cyst
  - Calcifying Odontogenic Cyst

- Non odontogenic/Pseudocysts
  - Nasopalatine duct cyst
  - *Nasolabial cyst*
  - *Thyroglossal duct cyst*
  - *Branchial Cleft cyst*
  - *Lymphoepithelial cyst*
  - *Dermoid cyst*
  - Simple bone cyst
  - Stafne defect

Radicular Cyst – Overview

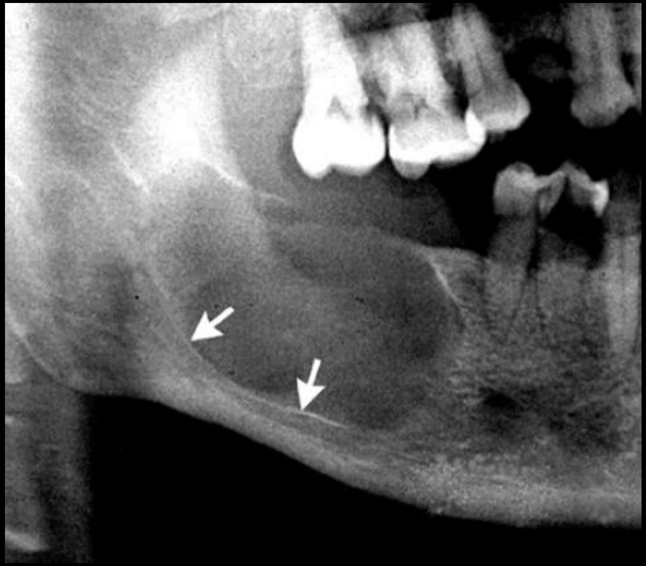
- Linked to periapical inflammatory disease
- Radiologic features overlap with abscess or granuloma
- Term used: **rarefying osteitis**
- Diagnosis confirmed via biopsy
- Lining derived from epithelial rests of Malassez
- Stimulated by inflammation from nonvital pulp

Residual Cyst – Overview

- Develops from incomplete removal of previous cyst lining
- Common sources: radicular cyst, dentigerous cyst
- Usually asymptomatic; found in edentulous areas
- May enlarge and cause painless swelling
- Pain possible if secondarily infected

Residual Cyst – Imaging & Management

- **Location**: Edentulous sites; more common in mandible
- **Periphery**: Well-defined, corticated unless infected
- **Internal**: Radiolucent; may show calcifications
- **Effects**:
  - Bone expansion and cortical thinning
  - Displacement of sinus floor or nerve canal
  - Tooth displacement or root resorption
- **Management**: Surgical removal or marsupialization



Dentigerous Cyst – Disease Mechanism & Clinical Features

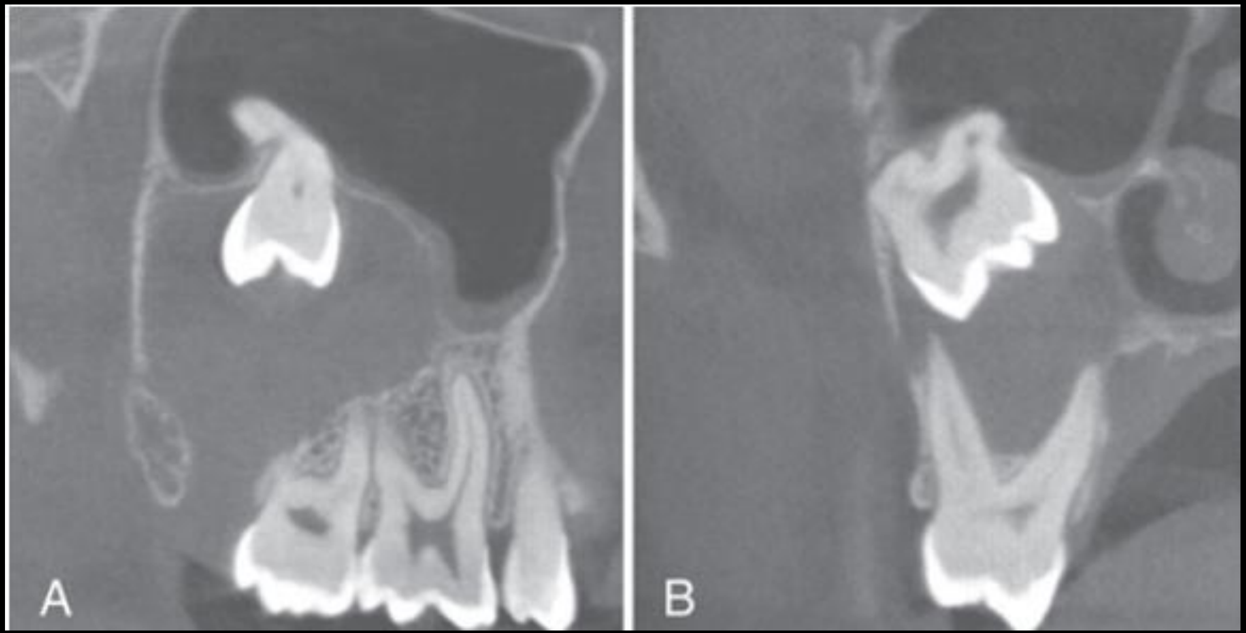
- Originates from reduced enamel epithelium
- Associated with impacted or supernumerary teeth
- Eruption cyst: soft tissue counterpart
- 2nd most common jaw cyst
- May present as missing teeth or facial asymmetry

Dentigerous Cyst – Imaging Features

- **Location**: Coronal to crown, often 3rd molars or maxillary canines
- **Periphery**: Well-defined, corticated; may lose cortex if infected
- **Internal**: Radiolucent except for tooth crown
- **Effects on structures**:
  - Displaces sinus floor, alveolar canal
  - Decompression reduces hydraulic contour

Dentigerous Cyst – Effects on Teeth

- Displaces and resorbs adjacent teeth
- Associated tooth may be displaced apically
- Maxillary teeth: displaced toward orbit
- Mandibular teeth: displaced into ramus, condyle, coronoid, or lingually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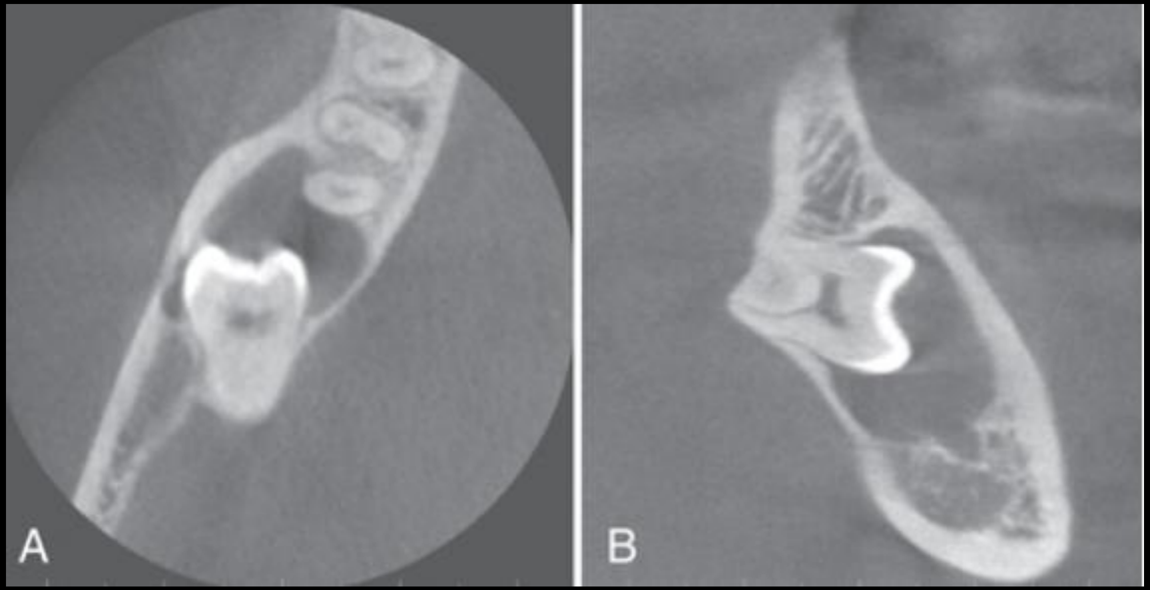
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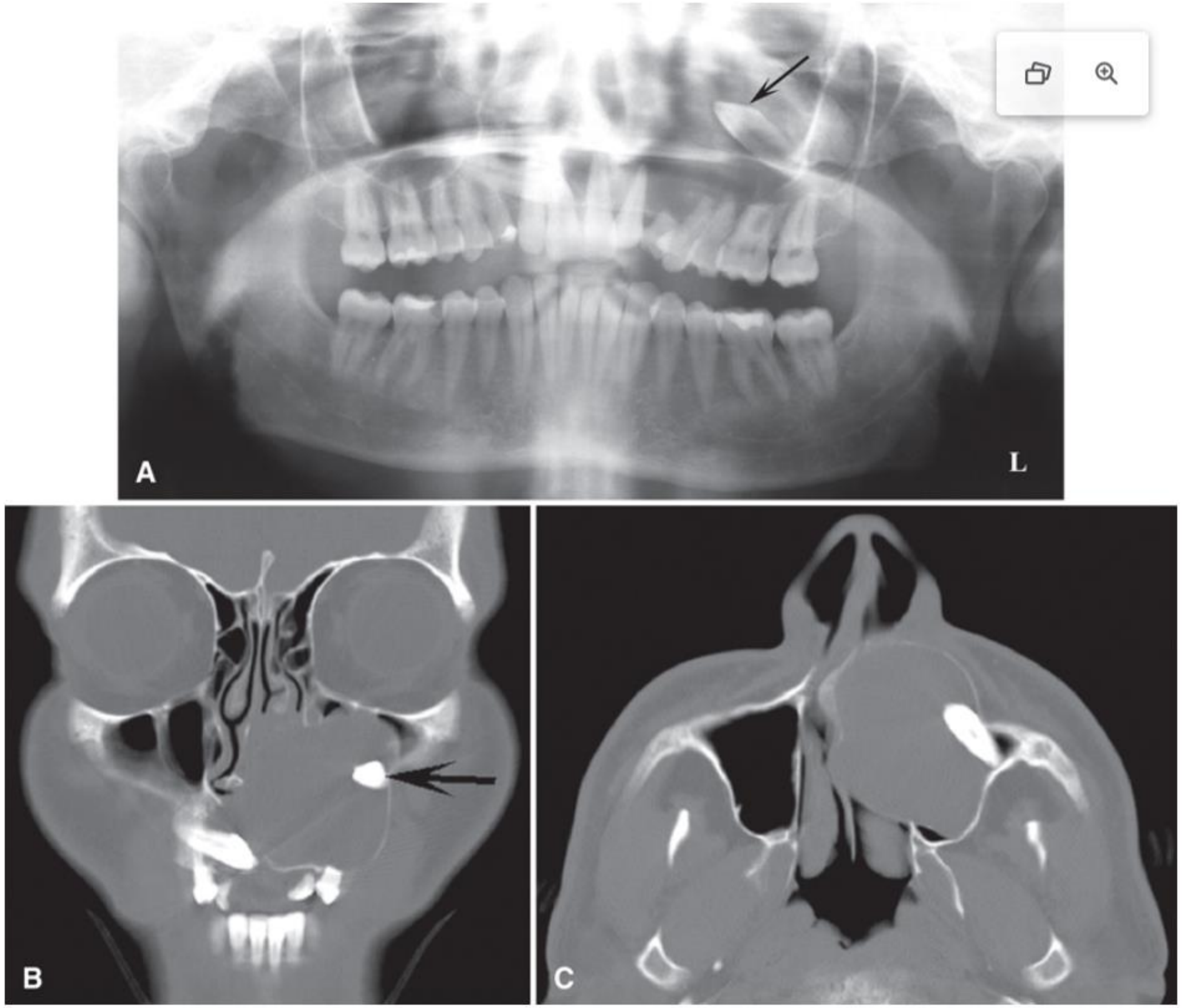
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Differential Interpretation of Dentigerous Cyst

- Histopathology alone is not diagnostic
- Diagnosis relies on:
  - Radiographic attachment at CEJ
  - Surgical observation
- Must rule out similar lesions:
  - Hyperplastic follicle
  - Odontogenic keratocyst (OKC)
  - Ameloblastic fibroma
  - Odontome
  - Ameloblastoma (unicystic or other)
  - Rare lesions: AOT, CEOT, COC. Rarefying osteitis may mimic dentigerous cyst

Management of Dentigerous Cyst

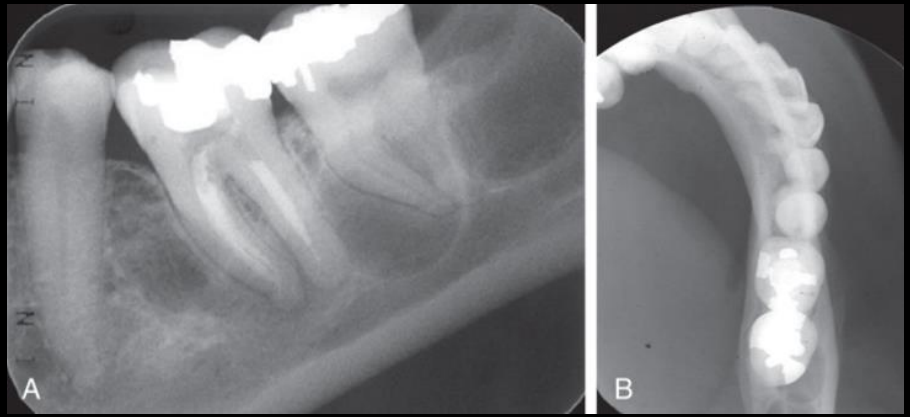
- Primary treatment: surgical removal (often includes tooth)
- Large cysts: marsupialization before excision
- Submit lining for histopathology
- Rare potential for malignant transformation:
  - Ameloblastoma
  - Squamous cell carcinoma
  - Mucoepidermoid carcinoma

Odontogenic Keratocyst – Disease Mechanism

- Origin: Dental lamina
- Lining: Thin, keratinized epithelium (4–8 cell layers)
- Keratin sloughing → cheesy material in lumen
- Growth potential due to **PTCH gene mutation**
- Bud-like epithelial proliferations → satellite microcysts
- Unique growth behavior and radiographic appearance

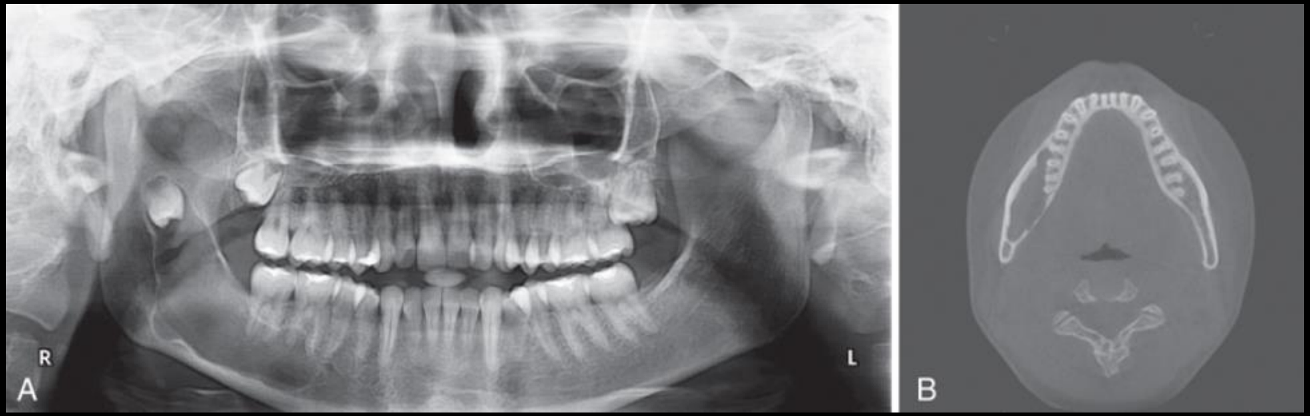
Clinical & Imaging Features of OKC

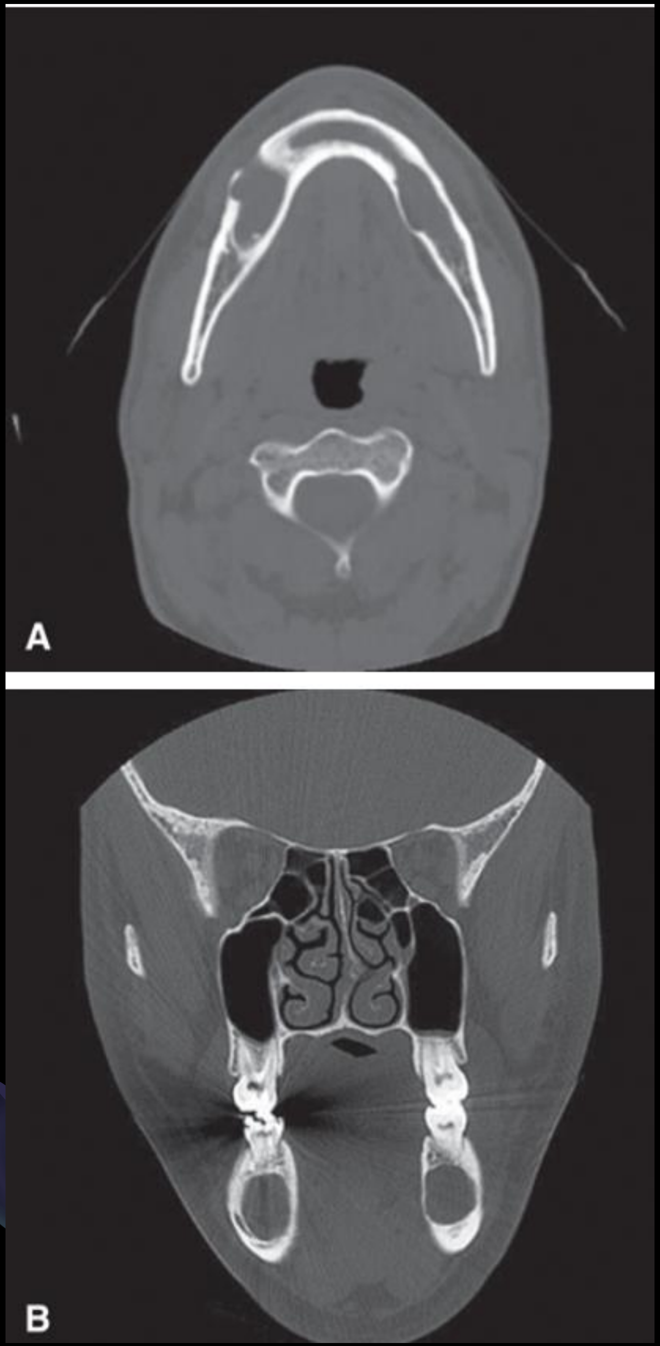
- Often asymptomatic; may cause mild swelling or pain if infected
- Aspiration: thick, yellow keratin material
- High recurrence rate due to residual epithelium
- **Location**: Posterior mandible, ramus; above inferior alveolar canal
- **Periphery**: Well-defined, corticated, may scallop cortex
- **Internal**: Radiolucent; may show keratin foci or septa (multilocular)



Effects & Differential Interpretation

- Grows **longitudinally** with minimal bone expansion
- May perforate cortex and contact soft tissue
- Displaces inferior alveolar canal
- May mimic:
  - Dentigerous cyst (if pericoronal)
  - Ameloblastoma (multilocular, more expansion)
  - Odontogenic myxoma
  - Simple bone cavity (SBC)
  - Lateral periodontal cyst

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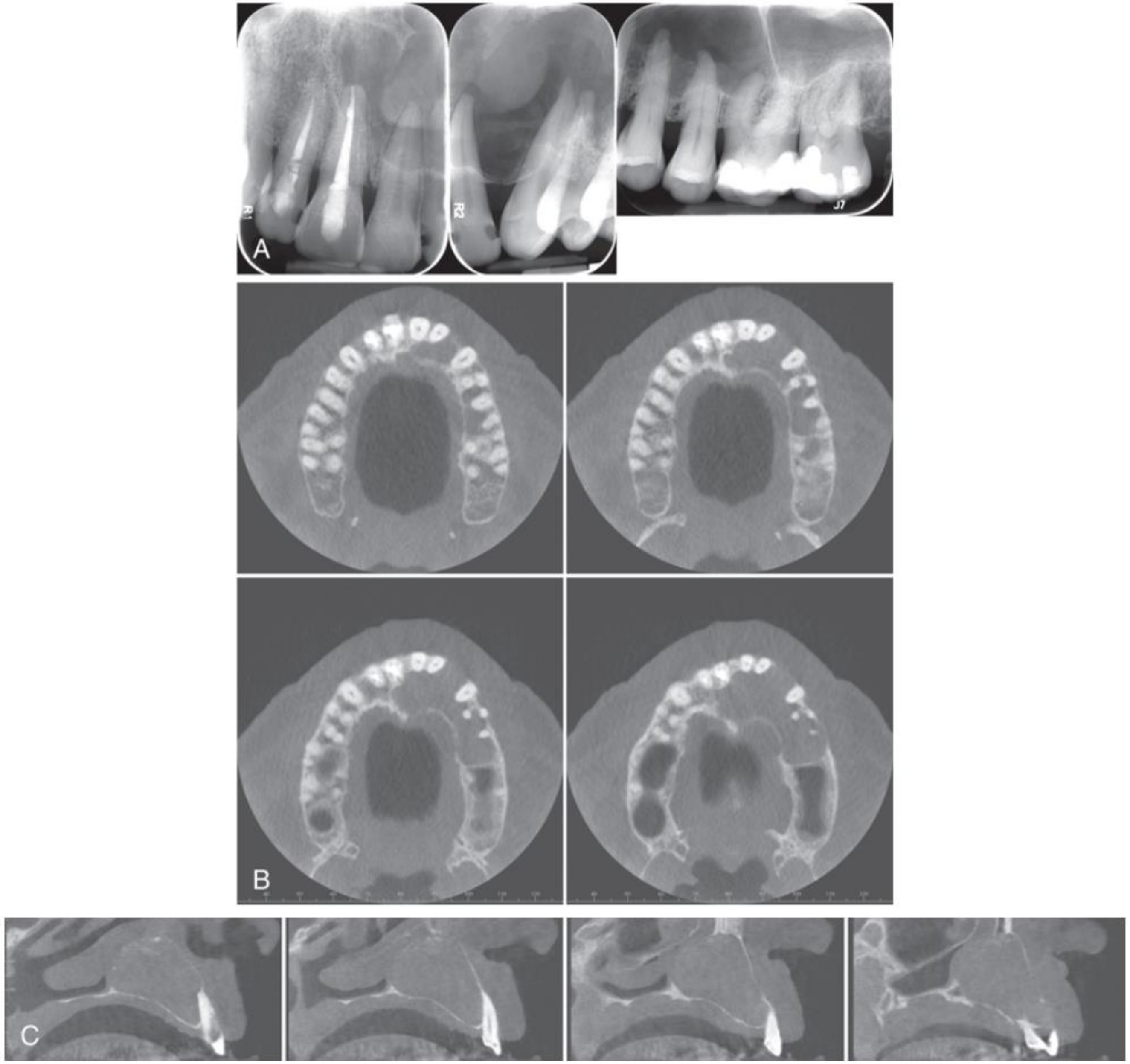
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Management of OKC

- Referral for full radiologic evaluation (CBCT or MDCT)
- MDCT preferred if cortical perforation suspected
- Treatment options:
  - Resection
  - Curettage
  - Marsupialization (for large lesions)
- Submit lining for histopathology
- Recurrence monitoring: follow-up for **5–10+ years**


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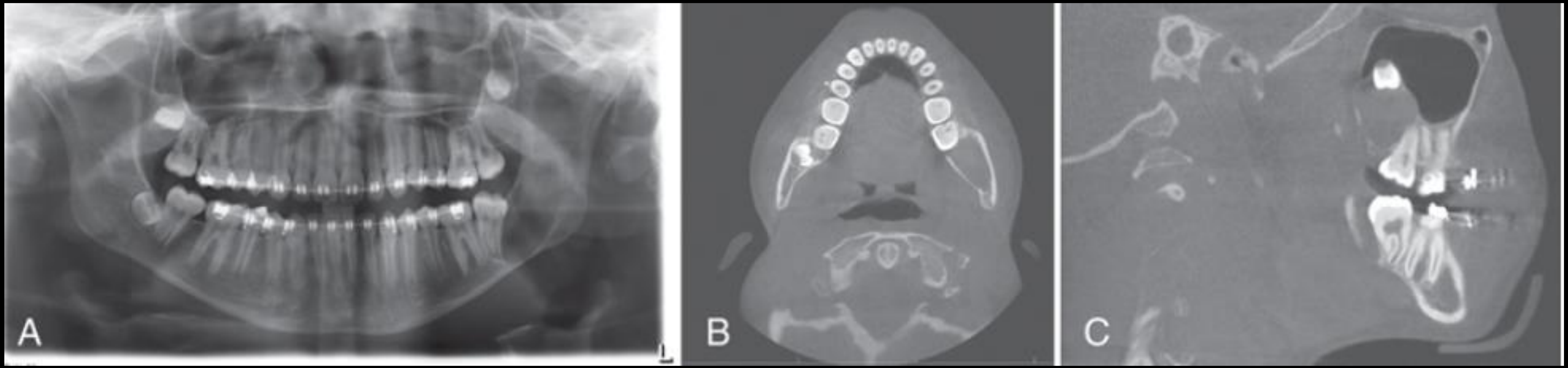
Nevoid Basal Cell Carcinoma Syndrome (NBCCS)

- Also known as **Gorlin-Goltz syndrome**
- Autosomal dominant; mutation in **PTCH gene**
- Variable expressivity:
    - Multiple basal cell carcinomas
    - Palmar/plantar pitting
    - Skeletal anomalies (bifid ribs, polydactyly)
    - Calcification of falx cerebri
    - Multiple OKCs
    - Medulloblastoma

NBCCS – Clinical, Imaging & Management

- Early detection: OKCs and skin lesions in 2nd–3rd decade
- 20% of cases from spontaneous mutations
- Imaging: CBCT/MDCT to detect multiple OKCs
- Differential diagnosis: Cherubism, multiple myeloma, dentigerous cysts
- Management:
  - Annual imaging follow-up
  - Referral to geneticist
  - Aggressive treatment due to high recurrence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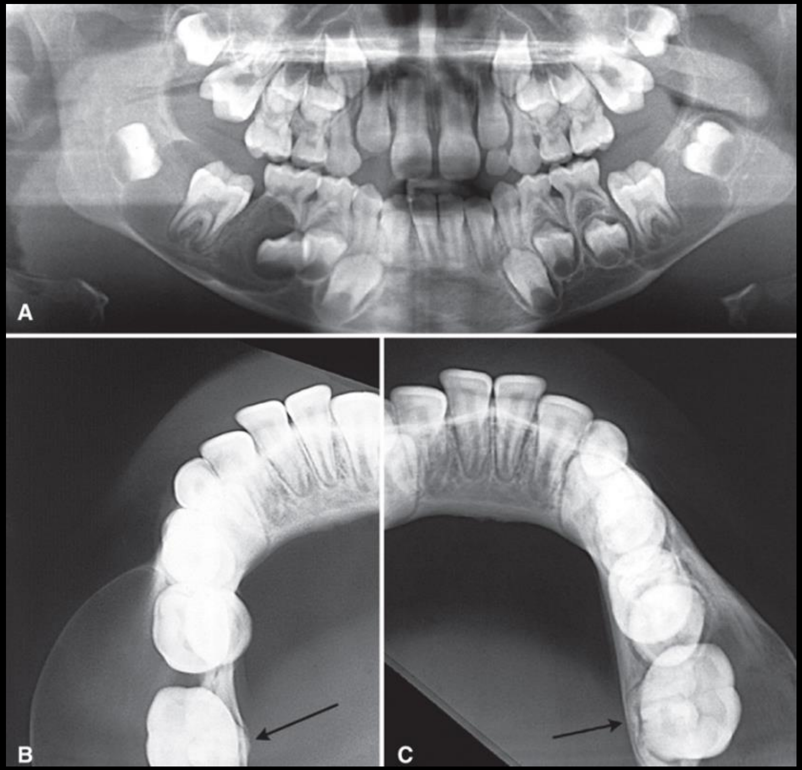
Buccal Bifurcation Cyst (BBC) – Overview

- Origin: Possibly epithelial rests of Malassez
- Common in **mandibular first molars**, vital pulp
- May present as:
  - Partially erupted molar
  - Lingual cusp tips protruding
  - Buccal hard swelling
- Age: Detected in first two decades



BBC – Imaging, Diagnosis & Management

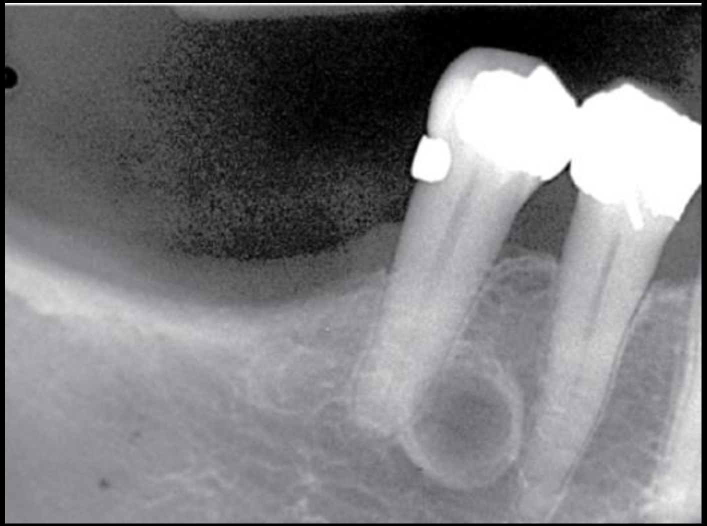
- **Location**: Buccal to molar furcation
- **Imaging**:
  - May appear subtle or well-defined
  - Best seen on cross-sectional occlusal image
- **Effects**:
  - Buccal bone expansion
  - Lingual tipping of roots, buccal tipping of crown
- **Differential**: OKC, dentigerous cyst, rarefying osteitis
- **Management**:
  - Conservative curettage
  - Tooth preservation
  - No recurrence

Lateral Periodontal Cyst – Overview

- Origin: Epithelial rests of Malassez in PDL space
- Usually unicystic; botryoid variant may appear multilocular
- Considered intrabony counterpart of gingival cyst in adults
- Typically small and asymptomatic
- May mimic lateral periodontal abscess if infected

Lateral Periodontal Cyst – Imaging & Management

- **Location**: Mandibular incisor/premolar area; maxillary lateral incisor/canine
- **Periphery**: Well-defined, corticated; round to oval
- **Internal**: Radiolucent
- **Effects**:
  - May efface lamina dura
  - Larger cysts can displace teeth



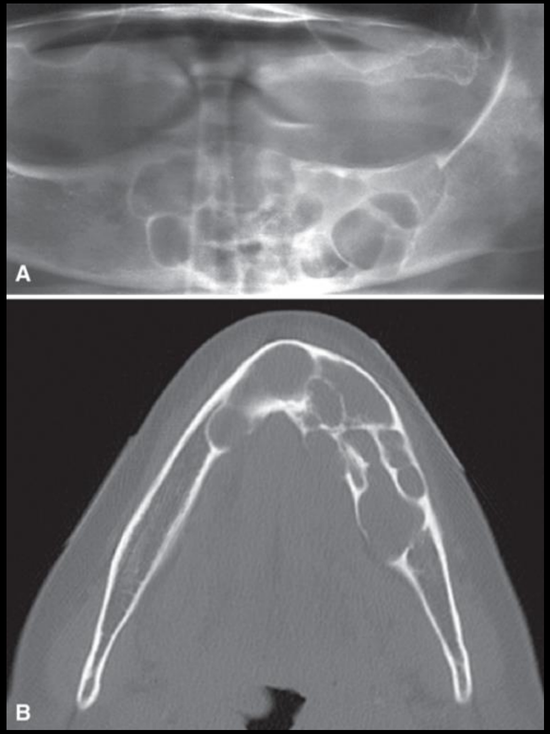
- **Differential**: OKC, SBC, ameloblastoma, rarefying osteitis
- **Management**: Excisional biopsy or enucleation; no recurrence

Glandular Odontogenic Cyst (GOC)

- Rare cyst from odontogenic epithelium
- May show salivary gland-like features (mucous cells)
- Possible link to central mucoepidermoid carcinoma
- Slight female predilection; average age ~50s
- Aggressive behavior and recurrence common

GOC – Imaging & Management

- **Location**: More common in anterior mandible and maxilla
- **Periphery**: Corticated; smooth or scalloped
- **Internal**: Radiolucent or multilocular
- **Effects**:
  - Bone expansion and cortical perforation
  - Tooth displacement
- **Management**: Aggressive surgical resection; regular follow-up imaging

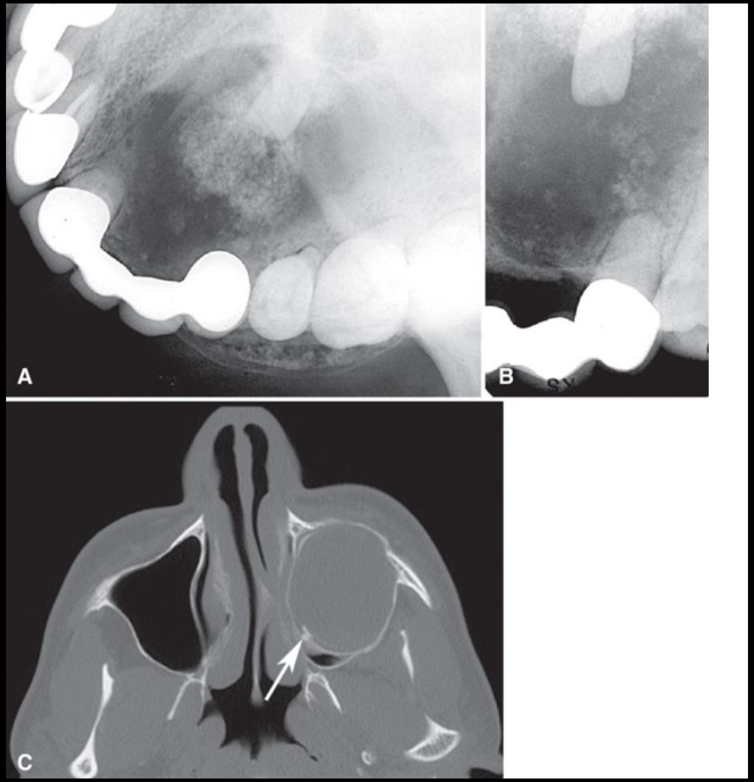


Calcifying Odontogenic Cyst (COC)

- Also known as Gorlin cyst, dentinogenic ghost cell tumor
- Uncommon, slow-growing, benign
- May resemble a cyst or solid neoplasm
- Can produce calcified matrix (dysplastic dentin)
- Sometimes associated with odontome

COC – Imaging & Management

- **Location**: Central in bone; often anterior to first molar
- **Periphery**: Corticated or irregular
- **Internal**:
    - Radiolucent or mixed with calcifications
    - May appear multilocular
- **Effects**:
    - Bone expansion and cortical perforation
    - Tooth displacement or root resorption
- **Management**: Enucleation and curettage; follow-up imaging for solid variants

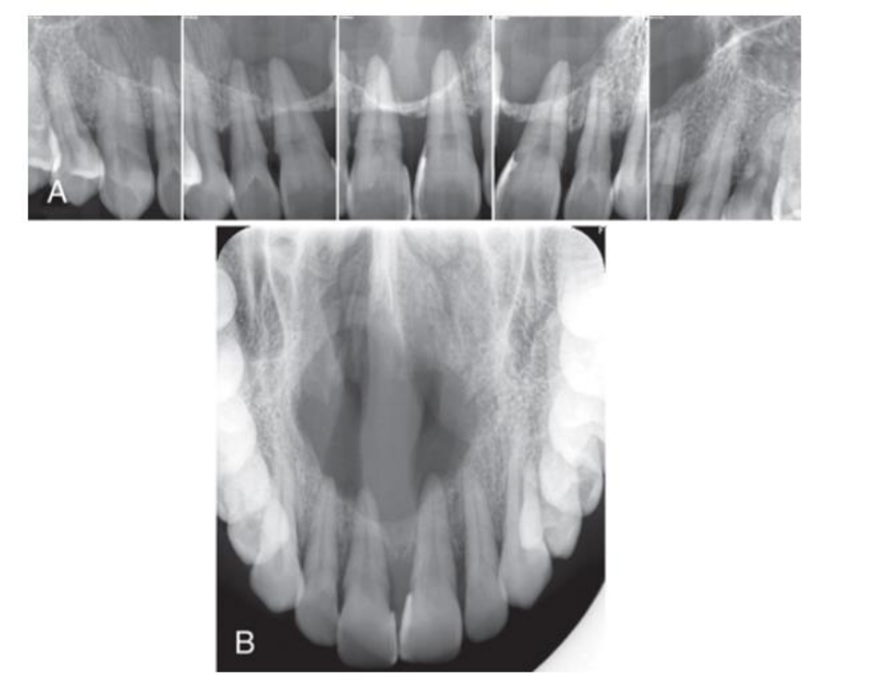


Nasopalatine Duct Cyst – Overview

- Originates from embryonic epithelial remnants in the nasopalatine canal
- Accounts for ~10% of jaw cysts
- Most common in males (3:1 ratio), ages 40–60
- Usually asymptomatic
- May present as:
  - Fluctuant swelling behind palatine papilla
  - Blue hue if superficial
  - Burning sensation or salty taste (nerve pressure or drainage)

Nasopalatine Duct Cyst – Imaging & Management

- **Location**: Maxillary midline, nasopalatine foramen/canal
- **Periphery**: Well-defined, corticated; may appear heart-shaped
- **Internal**: Radiolucent; occasional calcifications
- **Effects**:
    - Expansion of labial/palatal cortices
    - Displacement of nasal floor or central incisors
- **Differential**:
    - Large incisive foramen
    - Rarefying osteitis
- **Management**:
    - Enucleation (preferably palatal approach)
    - Marsupialization for large lesions

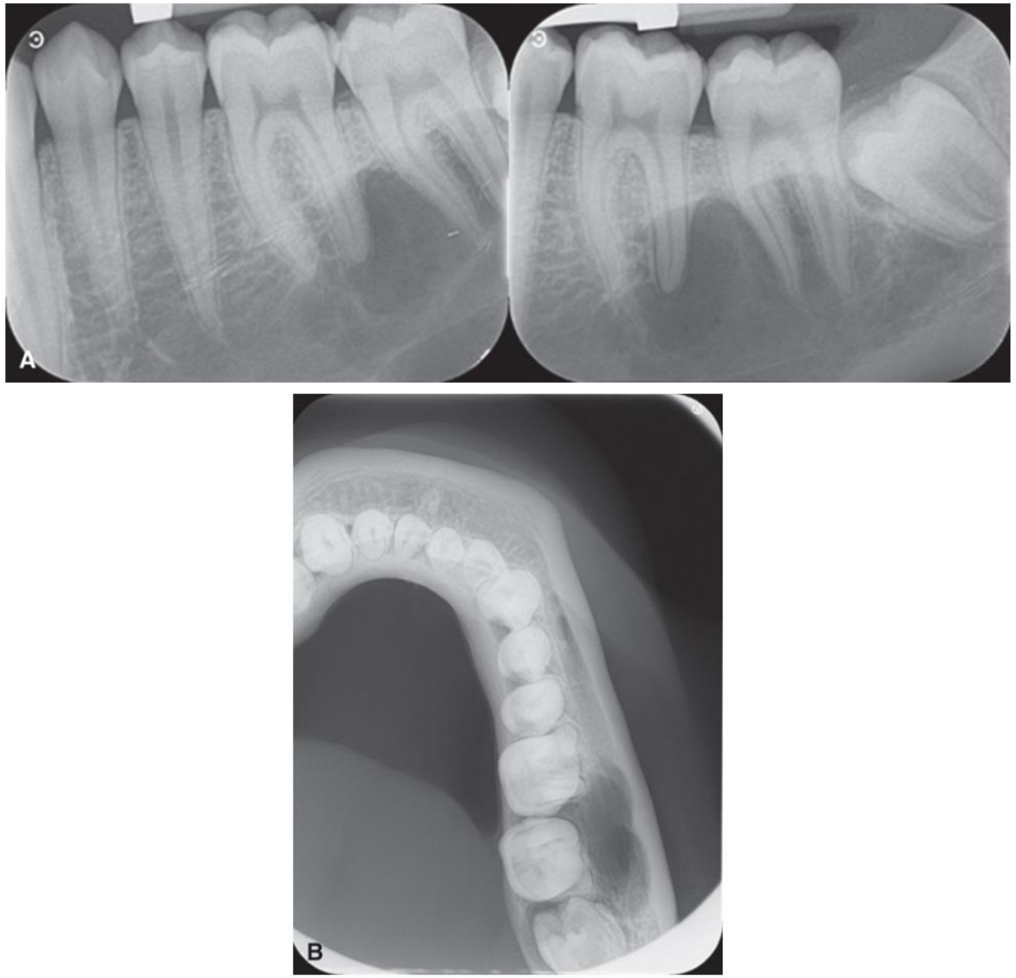


Simple Bone Cavity (SBC) – Overview

- Not a true cyst (no epithelial lining)
- Also known as: solitary, idiopathic, traumatic bone cyst
- May contain connective tissue, blood, or serosanguinous fluid
- Etiology: likely due to imbalance in osteoblast/osteoclast activity
- No evidence supporting trauma as a cause

SBC – Imaging, Features & Management

- **Location**: Mostly in posterior mandible; rare in maxilla
- **Periphery**: Well-defined, delicate cortex; scallops around roots
- **Internal**: Radiolucent; may show pseudosepta from scalloping
- **Effects**:
  - Minimal bone expansion
  - Rare tooth displacement or resorption
  - Teeth remain vital
- **Differential**: OKC, central giant cell lesion, malignancy (ruled out by intact lamina dura)
- **Management**:
  - Observation or cavity puncture
  - May heal spontaneously
  - No recurrence (unless associated with bone dysplasia)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Mandibular Lingual Bone Depression – Overview

- Also known as **Stafne defect**
- Not a true cyst: no epithelial lining or bone cavity
- Classified as a **pseudocyst** due to its cyst-like radiographic appearance
- First described in adults; not congenital
- Contents may include salivary gland tissue, fat, muscle, or lymph nodes

Stafne Defect – Imaging & Management

- **Location**:
  - Posterior mandible: below inferior alveolar canal
  - Premolar region: above mylohyoid ridge
- **Periphery**: Well-defined, corticated; cortex thickness varies
- **Internal**: Radiolucent
- **Effects**:
  - No impact on teeth
  - May thin lingual cortex; rarely contacts buccal cortex
- **Differential**:
  - OKC, central giant cell lesion (if cortex is thin)
- **Management**:
  - Recognition only; no surgery or advanced imaging needed